# Zijun Xiao

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Location: Atlanta, USA Highest Degree: M.S. Graduation: 2027.06
**U.S. Green Card Holder**

## Education

**2025.09 - 2027.06** **Georgia Institute of Technology**

Computer Graphics | M.S.

Focus: Real-time rendering pipeline, low-level graphics implementation, C++/Python, Unity development, machine learning, high-performance computing

**2024.09 - 2025.08** **Savannah College of Art and Design (SCAD)**

Game Development | M.S.

Focus: Unreal Engine game production, 3D art, art pipeline, DCC

**2019.09 - 2023.06** **South China Normal University**

Finance | B.S.

## Internship Experience

**2026.05 - Present** **Tencent · Game AI Engine Department** Technical Artist (PCG)

- **Agent×LLM×PCG Generation Pipeline:** Developed an Agent×LLM×PCG white-box auto-generation pipeline for the flagship title "Delta Force," enabling minute-level white-box level production—designers express intent in natural language while retaining fine-grained control via node editing.
- **Veda Front-end Interaction Tool:** Built Veda, a front-end interaction tool for level designers, supporting in-editor visual editing and real-time preview of PCG results inside UE5.
- **MochiAgent Natural-Language Orchestration:** Developed the MochiAgent natural-language orchestration module that translates designer intent into PCG node execution sequences, enabling conversational level design.
- **EasyLayout Pipeline Adaptation:** Adapted the EasyLayout generation pipeline to Mochi's unicast delegate interface, ensuring seamless integration between the generation workflow and the Agent orchestration system.
- **World-Model Content Pipeline (One-Click "Params → Video" Batch Production):** Enter fps / count / duration (optional theme) in the web UI; the pipeline autonomously runs LLM prompt generation → AI three-image set → UE white-box scene → automatic camera rig → MRQ rendering → mp4 export, with real-time progress visualization and in-page video playback. Six modules (front-end / prompt-gen / Phase A images / Phase B white-box capture / Phase C encoding / progress aggregation) form a closed end-to-end data-flow loop.
- **Core Mechanisms:** Dynamic capacity—defines real throughput by surviving camera rigs per scene and consumes the scene pool on demand to avoid wasted image-API calls; Parameter injection chain—fps×duration propagates through CLI → capture_config → MRQ/mp4 frame rates for full consistency; Resumable + circuit breaker—idempotent stages, per-shard fresh editor processes, MRQ timeout drain, hard disk-peak precheck. Validated by 25-agent adversarial review; all 22 confirmed defects fixed.

**2023.07 - 2023.12** **Seasun (Kingsoft)** Technical Artist

- **Engine Upgrade & Rendering Validation:** During the core project's engine migration, participated in cross-engine art asset migration and conducted compatibility testing and visual validation of the new rendering pipeline to ensure smooth transitions of complex scenes and material effects.
- **Performance Profiling & Optimization:** Used performance monitoring tools for rigorous rendering stress tests on high-load scenes; helped locate and resolve low-level asset issues including excessive Draw Calls, abnormal Overdraw, and VRAM over-budget, significantly reducing in-game frame drops and stutter.

- **Technical Art Audit:** From a performance-optimization perspective, helped establish art asset production specs for the new engine and built a pre-emptive asset inspection flow that strictly controls performance budgets per stage while maintaining high visual quality.

## Research Experience

**2025.12 - Present** **Frontier Lab (Tencent)** Next-Generation Neural Rendering Research

Tencent Game AI Engine Dept. · Frontier Rendering Group — Participated in "Exploring Next-Generation Neural Rendering," focused on deploying neural rendering as a deployable, interactive, editable, and generative game-world representation.

- **Explicit 3D World Model:** Upgraded 3DGS from "viewable visual reconstruction" to a "deployable, interactive, editable, generative" explicit world model for game scenes; designed large-scale tiled capture and reconstruction workflows covering complex regions based on engine-controlled cameras, GT pose, depth, Mesh, and semantics.
- **Lightweight Mobile 3DGS Rendering:** Lightweight 3DGS rendering for mobile/Web — dynamically balancing quality and frame rate via block / LoD tree / streaming / compression / fixed budget; co-scheduling Mesh and 3DGS by LoD level to resolve depth occlusion, boundary blending, and lighting consistency for stable on-device deployment.
- **Transformer-based & Generative Neural Rendering:** Explored Transformer-based Rendering (Scene tokens → Image tokens, Attention modeling light transport); transformed low-cost G-buffer / low-poly renders via upscaling + style transfer + GI completion + temporal synthesis into high-quality, style-/semantic-controllable final images.
- **Engineering Deployment & Closed-loop Validation:** Developed frontier complex materials on the UE5 Substrate architecture with custom material attributes for Neural Rendering; independently built a C++ UE5 editor plugin to auto-extract and clean massive asset data, feeding datasets to neural rendering models; completed closed-loop validation under real engine constraints of frame rate / VRAM / bandwidth / power.

## Project Experience

**2022.12 - Present** **Personal Projects**

**ARPG Game Project | Technical Artist / Technical Designer**

- **Skill VFX & Visual Feedback:** Implemented high-precision combat hit detection; developed complex skill-effect shaders based on vertex offset (WPO) and dynamic masks, strengthening the impact and visual tension of core combat.
- **Scene Performance Optimization & Memory Tuning:** Led performance management for high-load open-world scenes. Applied multi-level LOD strategies, strict static/dynamic batching (GPU Instancing / Batching), and memory culling. Reduced overall rendering overhead by 30% without quality loss.
- **AI Logic & Multi-Entity Performance Management:** Combined Behavior Tree and FSM to build multi-stage hardcore Boss fights; optimized CPU Tick cost for entity-combat scenarios, ensuring absolute frame-rate stability for complex AI computation.

**Parkour Animation & DCC Tooling System | Technical Animation**

- **Procedural Animation & IK Solver:** Used inverse kinematics (IK) and engine animation blueprints to solve bipedal ground-conforming on complex uneven terrain and procedural climbing hand interaction, greatly improving physical realism and coherence of dynamic motion.
- **DCC Automation Pipeline & Standards:** Wrote Maya automation scripts in Python/MEL, enabling one-click character and animation asset import from DCC to UE5 (Pipeline). Established and enforced rigorous skeleton and rigging standards for efficient, seamless art-asset iteration.

## Skills

- **Programming Languages:** C++, C#, Python, Java, HLSL
- **Engines & Tools:** Unreal Engine 5, Unity, Maya, Blender, Substance Painter/Designer, RenderDoc

• **Core TA Domains:** Custom rendering pipelines, performance profiling & memory management, DCC automation pipelines, complex materials & shader development, procedural animation (IK), 3DGS Gaussian Splatting, PCG / Agent automation pipelines
• **R&D Capabilities:** Cross-functional collaboration (game design / programming / art), automated asset review & massive data processing, cutting-edge technology engineering deployment